

Shuai Li

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Education

University of Florida	Geography (Climate Modeling and Data Fusion)	Ph.D., 2024-present
University of Wyoming	Ecology (Ecosystem Science and Management)	Ph.D., 2023-2024
Chengdu University of Technology	Geography (Human Settlements and Urban Rural Planning)	M.S., 2022
Zhengzhou Normal University	Geography Science	B.S., 2019

Academic Appointment

Research Assistant GeoDi Lab, Department of Geography, University of Florida	2024 - present
Research Assistant GeoDi Lab, Department of Geography, University of Wyoming	2023 - 2024

Research Experience/Projects

NASA EPSCoR

Quantifying Effects of Land Cover Change-Climate Interaction on Ecosystem Productivity Over Western North America

Wyoming Citizen Science Data Clearinghouse

In charge of the backend development of the clearinghouse platform

Philosophy and Social Sciences Research Fund of Chengdu University of Technology (YJ2021-YB023)

Research on the construction of ecological security pattern in Chengdu-Chongqing urban agglomeration

Key Laboratory Project of Beijing Bamboo Science and Technology Open Fund (ICBR-2020-16)

Study on carbon and water utilization efficiency of southern bamboo forest ecosystem

Sichuan Provincial Department of Natural Resources Project (FZC2020004-6)

Risk investigation and evaluation of geological hazards in Pengzhou City, Sichuan Province

Key Research Project of National Park Research Center (GJGY2019-ZD003)

Simulation and prediction of ecological landscape pattern evolution in giant panda National Park

Hebei Key Laboratory of Wetland Ecology and Protection Open Fund (hklk201902)

Study on wetland ecological process in Beijing Tianjin Hebei region under the action of environmental geology

Geological Survey Project of Sichuan Province (DD20190542)

1:250000 ecological and geological survey of Mianning County, Liangshan Prefecture

Publications In Prep/Under review

Li, S., Yang, D., He, Y., Li, Y., & Liu, W. A Geospatial Foundation Model Driven Framework for Remote Sensing Spatiotemporal Fusion. *GIScience & Remote Sensing*. Under Review.

Li, S., Yang, D., He, Y., Li, Y., & Parazoo, N. Hydroclimatic Responses to Land Cover Change in the Arctic-Boreal Region: Separating Direct Land Surface Effects from Land-Atmosphere Feedbacks. In Prep.

Li, S., Yang, D., He, Y., Keellings, D., & Silva, C.A. Assessing the Impacts of Land Use and Land Cover Change on Regional Climate in the Edmonton Region. In Prep.

Li, S., Yang, D., & Fill, J. Earth Observation-Based Investigation of the Thermal Dynamics in Longleaf Pine Ecosystems. In Prep.

Li, Y., Li, S., Yang, D., & Liu, W. Wildland-Urban Interface (WUI) Vulnerability Assessment: 1990-2020 Temporal Analysis at California NEON Sites and Surrounding Areas. In Prep.

Book Chapter

Yang, D., Tai, X., He, Y., Liu, T., Li, S., & Katabuchi, M. (2026). Deforestation assessment. *Data-Driven Earth Observation for Disaster Management: From Theory to Practical Applications*, December 2025, 347-367; Elsevier, ISBN: 978-0-443-33803-8, DOI: 10.1016/C2024-0-00164-6.

Publications

He, Y., Yang, D., Li, S., Cheng, L., & Fang, F. (2026). Enhanced Extreme Precipitation Simulation in China using NCAR CESM Based on a Realistic Remotely-Sensed Time Series of Annual Land Cover and Land Use Data from 1982 to 2013. *Earth and Space Science*, 13(6), e2025EA004618.

Li, S., Yang, D., He, Y., Parazoo, N., & Liu, W. (2026). The impact of land cover change-climate interactions on ecosystem productivity in the Arctic-Boreal region. *Agricultural and Forest Meteorology*, 383, 111137.

Li, S., He, Y., Yang, D., Li, Y., & Liu, W. (2026). Incorporating learned geospatial embeddings to deep image prior for inpainting cloud areas in remotely sensed images. *Science of Remote Sensing*, 100404.

Li, S., Yang, D., He, Y., Parazoo, N., & Liu, W. (2025). The biogeophysical impacts of land cover change on climate extremes in the Arctic and Boreal regions. *Environmental Research Letters*, 20(8), 084057.

Liu, W., Li, S., Fan, D., Wen, Y., Madson, A., Mitchell, J., He, Y., & Yang, D. (2025). A Deep Learning Workflow for CORONA-based Historical Land Use Classifications. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*.

Li, S., Ma, H., Yang, D., Hu, W., & Li, H. (2023). The Main Drivers of Wetland Evolution in the Beijing-Tianjin-Hebei Plain. *Land*, 12(2), 480.

Li, S., Ni, Z., Zhao, Y., Hu, W., Long, Z., Ma, H., Zhou, G., Luo, Y., & Geng, C. (2022) Susceptibility Analysis of Geohazards in the Longmen Mountain Region after the Wenchuan Earthquake. *International Journal of Environmental Research and Public Health*, 19(6), 3229.

Liu, Y., Ni, Z., Zhao, Y., Zhou, G., Luo, Y., Li, S., Wang, D., & Zhang, S. (2022) Spatial-Temporal Evolution and Driving Forces of Drying Trends on the Qinghai-Tibet Plateau Based on Geomorphological Division. *International Journal of Environmental Research and Public Health*, 19(13), 7909.

Xiong, Z., Ma, H., Li, S., & Zhang, N. (2022). Summary of the Application and Prospect Analysis of Knowledge Graphs in the Marine Field. *Computer Engineering and Applications*, 58, 15-33.

Conference Presentation

Li, S., He, Y., Yang, D., Li, Y., & Liu, W. Leveraging Semantic Priors from Learned Geospatial Embeddings to Enhance the Deep Image Prior for Inpainting Cloud Areas in Remotely Sensed Imagery. ForestSAT 2026, Gainesville, Florida, USA. Oral Presentation.

Li, S., He, Y., Yang, D., Parazoo, N., & Liu, W. The Biogeophysical Effects of Land Cover Change on Climate Extremes in the ABoVE Domain. 2025 AAG Annual Meeting, Detroit, Michigan, USA. Oral Presentation.

Li, S., Yang, D., He, Y., & Parazoo, N. Land Cover Change-Climate Interactions Amplified the Diminishment of Spring Ecosystem Productivity in the Arctic-Boreal Region. AGU24 Annual Meeting, Washington, D.C., USA. Oral Presentation.

Li, S., Yang, D., & He, Y. Regional Climate Effects of Long-term Land Cover Change: A Case Study in the ABoVE Domain. 2024 AAG Annual Meeting, Honolulu, Hawai'i, USA. Oral Presentation.

Yang, D., Li, S., He, Y., & Parazoo, N. Counteracting Climate Change: How Land Cover Shifts Regulate Temperature in the Vulnerable ABoVE Domain. 11th International Carbon Dioxide Conference (ICDC11), Manaus, Amazonas, Brazil. Poster Presentation.

Research Grants

Co. P.I.	NCAR Wyoming Supercomputing Center, University of Wyoming, Gong, J., Li, S. , Yang, D., 2026-2027 (500,000 Credit Hours)
Lead. P.I.	NCAR Wyoming Supercomputing Center, University of Wyoming, Li, S. , 2024-2025 (1,000,000 Credit Hours)
Co. P.I.	NCAR Wyoming Supercomputing Center, University of Wyoming, Yang, D., Li, S. , 2024-2025 (1,000,000 Credit Hours)
Co. P.I.	NCAR Research Computing Grant, University Corporation for Atmospheric Research, Yang, D., He, Y., Li, S. , 2023-2024 (610,000 Credit Hours)

Awards

2024-2026	Graduate Student Geography Department Travel Grant, University of Florida	\$1,200
2024	Graduate Travel Awards, University of Florida	\$1,000
2024	GPSC Travel Grants, University of Arkansas	\$1,000
2023	NASA EPSCoR Travel Grants, NASA Wyoming Space Grant	\$500
2021	Academic scholarship, Chengdu University of Technology	¥ 6,000
2020	Academic scholarship, Chengdu University of Technology	¥ 6,000
2020	Excellence Award, Chengdu University of Technology	

Undergrad Mentorship

WUI Vulnerability Paradox: Development Patterns and Fire Risk	2025
Cross-Boundary Fire Management: When Ownership Meets Ecology	2025

Teaching Experience

EMERGE NASA Data Hackathon

- Geospatial statistics using GEE

Google Earth Engine Workshop

- Overview of GEE
- Basic operation and application of GEE

Service

2026 – Present	Reviewer: Journal of Geophysical Research - Biogeosciences
2024 – Present	Reviewer: Scientific Reports

Technical Expertise

- Programming: Python, R, MATLAB,
- Data Visualization/Software: ArcGIS, GEE (JavaScript), ENVI,

Professional Organizations

- American Geophysical Union (AGU)
- Association of American Geographers (AAG)